

Mode Collapse

Goodfellow et al., “Generative Adversarial Networks” — abstract, Figure 1, §3

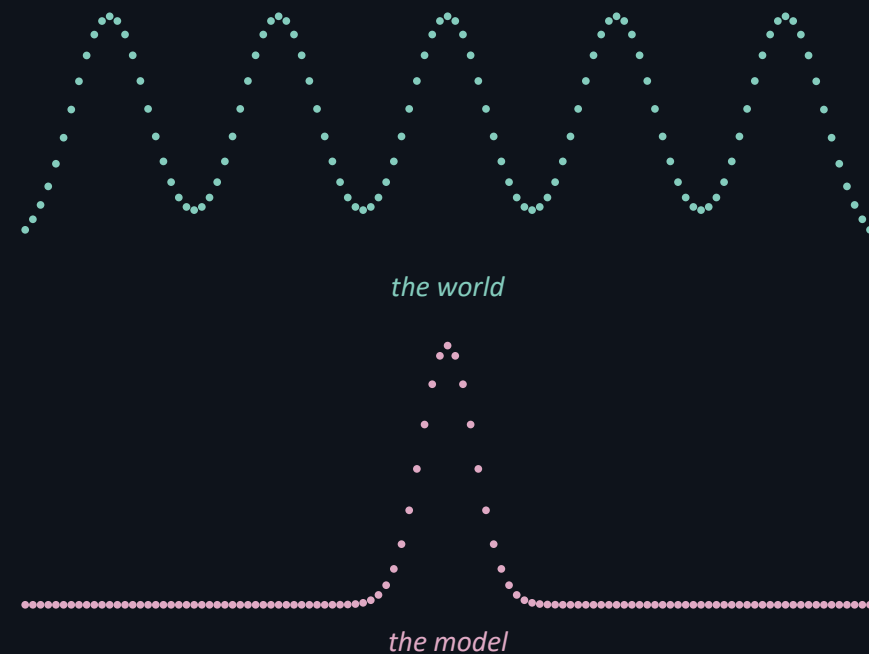
Bender et al., “On the Dangers of Stochastic Parrots”

Ruha Benjamin, Race After Technology — excerpt

Tier 2 exercise: forty variations on one prompt. Bring the results.

Session length: 5 hours, with four breaks

Key question: what disappears when a system optimises for approval?



New symbols this week — last week's all still apply

p

a distribution: how often each possible thing happens. $p(x)$ is how common x is.

p_data vs p_g

the real distribution of the world (data) versus the model's distribution (generated).

G

the generator. A function that turns random noise into an output.

D

the discriminator, or critic. A function that scores how real something looks, 0 to 1.

z

random noise fed into G. The only source of variety the generator has.

E[·]

the average of the thing in brackets, taken over many samples.

KL(p||q)

how badly q fails as a stand-in for p . Zero only if they are identical.

JSD(p||q)

a symmetric, well-behaved cousin of KL. Also zero only if identical.

H(p)

entropy: how spread out a distribution is. High means varied, low means concentrated.

min max

a game: one player is trying to make the number small, the other large.

In the papers these carry subscripts — E is written with the distribution underneath it. The subscript only says which pile of samples you are averaging over.

Write before we begin

Describe a time you learned what a particular audience approved of — a teacher, a workshop, an editor, a platform, a family — and began producing that thing.

Then write down what you stopped making. Not what you felt about it. What specifically stopped coming out.

Keep the page.

● KEY QUESTION FOR THE WEEK

What disappears when a system optimises for approval?

Note that the question does not ask what goes wrong. A collapsed generator has not broken. It is succeeding — narrowly and perfectly — at the thing it was asked to do. The diversity of the world was never part of the objective, so the diversity of the world quietly disappears.

The word “system” is doing a lot of work in that sentence. Holding it open is this week’s discipline.

The unit has two jobs, and they pull against each other

1 Take the metaphor seriously

A generative model that has found the small set of outputs which reliably satisfy its critic, and produces those and only those, forever, is an extraordinarily precise description of something that also happens to writers, to institutions, and to cultures. Sit with how good it is.

2 Then be suspicious of it

The temptation to reach for a technical term as a poetic figure is exactly the intellectual move this course is training you to make carefully, with the mechanism in hand. A metaphor that fits too well is usually doing less work than it appears to.

Which is why the mathematics comes first

You cannot be responsibly suspicious of a technical metaphor you do not understand. So we spend two hours on the actual mechanism — the game, the objective, the reason a perfectly working optimiser collapses — and only then ask which parts of it transfer to a person, an institution or a culture, and which parts do not. By 4:30 you should be able to say exactly where the analogy stops carrying weight.

PART ONE

Your forty outputs

First look, before any theory · 0:20–0:40

Describe before you diagnose.



Lay them out, and answer four questions in plain language

1 What is the single most common thing?

Not a theme — a specific recurring element. A name, a setting, a sentence rhythm, a colour, a structure. If more than one of you generated the same phrase, say so out loud now.

2 How many genuinely different outputs are there?

Count them by hand. Two outputs that differ only in a swapped adjective are one output. Agree on the rule with your neighbour before you count, and write the rule down.

3 What did you expect that never appeared?

This is the hard one and it is the whole exercise. List things that plainly belong to the space of the prompt and are entirely absent across all forty.

4 Whose absence would you not have noticed?

Something missing that you only see because someone else pointed it out, or that you had to reason your way to rather than feel. Mark those differently.

No technical vocabulary in this block. Not mode, not collapse, not distribution. Describe what is on the page in the words you had before this course started.

A prediction you can check against your own pages

Names and settings converge hard

A handful of names recur across dozens of outputs, often across different students. Settings cluster in a narrow band. This is the most visible layer and the easiest to notice.

Structure converges harder

Sentence length, paragraph shape, where the turn happens, how things end. Students notice this second and find it more unsettling, because they had not been watching for it.

Stance converges hardest

Tone, degree of resolution, whether anything is left unresolved, who gets sympathy. This layer is nearly invisible until someone names it, and it is the one that matters most for the afternoon.

And one finding that is not about the model

Some of the convergence is yours. Forty variations on one prompt, written by one person in one sitting, are forty outputs from a system that includes you — your prompt, your idea of what counts as a variation, your decision to stop. Hold that. We will need it at 4:30, when the question is whether the same word describes both halves of that system.

PART TWO

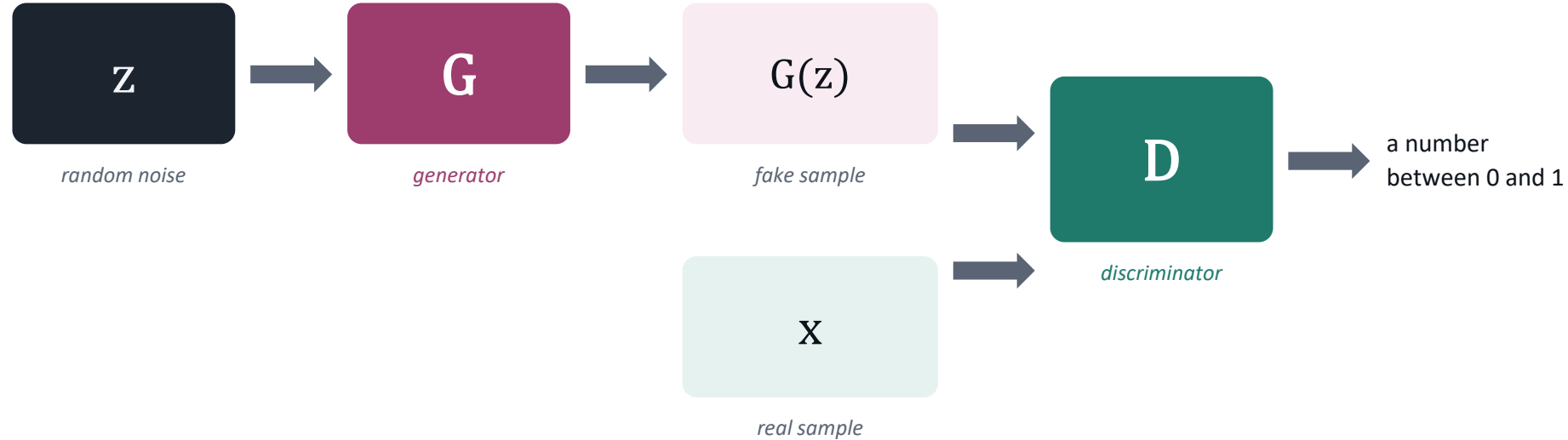
The adversarial game

Goodfellow et al. §3 · 0:40–1:30

Two networks, one objective, and a minus sign between them.



Two networks with opposite jobs



D wants to be right

Output near 1 for real samples, near 0 for generated ones. It is a classifier, trained exactly as last week: a loss, a gradient, a step. Nothing new.

G wants D to be wrong

It never sees the real data. Its only signal is what D says about its own output. It succeeds when D cannot tell the difference.

And they train together

Step D, step G, step D, step G. Each player's loss depends on the other's current state, so the landscape from last week is now moving under your feet.

The objective, written down

G tries to make this small · D tries to make it large

$$V(D, G) = E [\log D(x)] + E [\log(1 - D(G(z)))]$$

as printed: min over G, max over D, of V(D,G) — the first average over real data $x \sim p_{\text{data}}$, the second over noise $z \sim p_z$

“D tries to make this number big. G tries to make it small. They are handed the same number and told opposite things.”

V(D, G)

one number, the value of the game. It depends on both players at once.

E [·]

average over many samples. Everything here is an average, never a single case.

log D(x)

big when D correctly says a real sample is real. D pushes this up.

1 – D(G(z))

big when D correctly says a generated sample is fake. D pushes this up too.

min / max

the minus relationship. Whatever helps one player hurts the other, exactly.

Look for a term that rewards variety. There is not one. Nothing in V measures how many different things G produces, and nothing in V goes up when G covers more of the world. That absence is not an oversight on this slide — it is the whole unit.

A practical fix, buried in one paragraph

The problem the authors name

Early in training G is bad, so D rejects its output confidently. Then $\log(1 - D(G(z)))$ is nearly flat: the gradient is tiny and G barely learns. The objective is correct and useless.

So they change what G maximises

G maximises $E [\log D(G(z))]$

instead of minimising $E[\log(1 - D(G(z)))]$. Same fixed point, much stronger gradient early on.

Why this matters for today

Under the new objective G is rewarded purely for how convincing its output is. Not for how many different convincing things it makes. Any single very convincing output scores maximally.

A change of convenience

It was adopted because the first version trained badly. It is a practical repair, described in a paragraph, and it changed the character of the failure mode.

This is last week's η again

A choice made for tractability, not stated as a value judgement, permanently shaping what the artefact turns out to be. Watch for these; they are everywhere and they are rarely documented.

And the paper is honest about it

Goodfellow et al. name the saturation problem and the workaround openly. The failure is not concealment. It is that a footnote-sized engineering decision has consequences nobody sized at the time.

What the perfect critic would say

$$D^*(x) = p_{\text{data}}(x) / (p_{\text{data}}(x) + p_{\text{g}}(x))$$

“The best possible critic reports the share of the traffic at x that is real.”

$p_{\text{data}}(x)$

how often the real world produces x .

$p_{\text{g}}(x)$

how often the generator produces x .

the ratio

if only real data lands at x , $D^* = 1$. If only fakes land there, $D^* = 0$.

$D^* = 1/2$

what you get when $p_{\text{data}} = p_{\text{g}}$ everywhere: the critic is reduced to a coin flip.

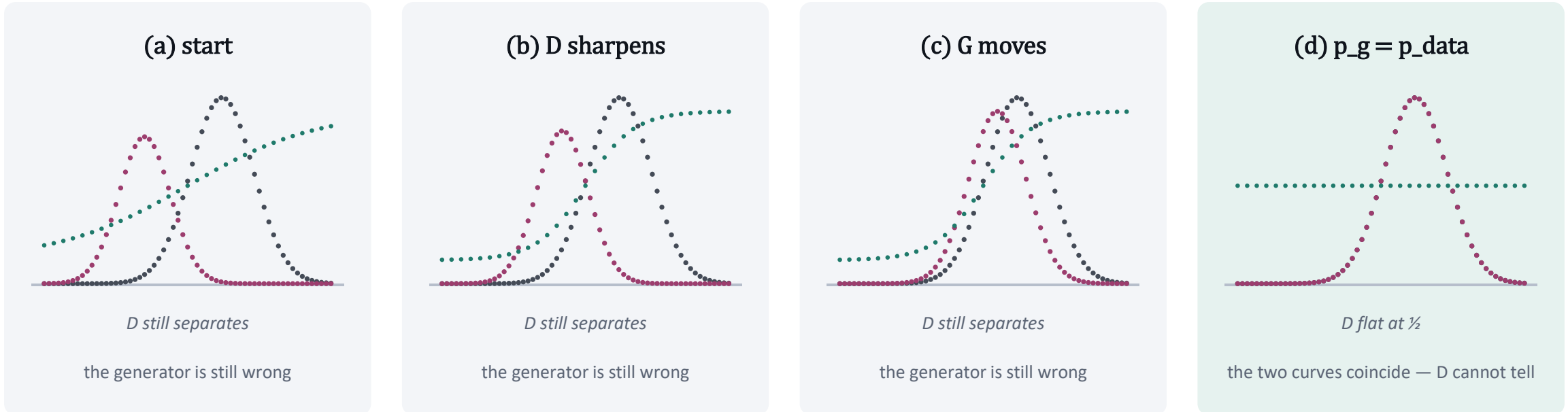
$C(G) = -\log 4 + 2 \cdot \text{JSD}$

substituting D^* back in, the game's value is a distance between the two distributions.

What the theory promises, and what it does not

The theorem says that if you could solve the game exactly, the unique optimum is $p_{\text{g}} = p_{\text{data}}$ — the model reproduces the world, mode for mode. That is a real and beautiful result. It is also a statement about an idealised optimum, not about what alternating gradient steps on two finite networks will actually find. The gap between those two sentences is where mode collapse lives.

What convergence is supposed to look like



grey = the real distribution p_{data} · magenta = the generator p_g · teal = what the discriminator D outputs

Notice what the figure assumes

The paper's illustration uses a single-humped world. Under that assumption a generator that finds the hump has found everything, and convergence and coverage are the same event. The figure is honest and it is also the best possible case. Redraw the world with five humps and the picture stops being reassuring.

PART THREE

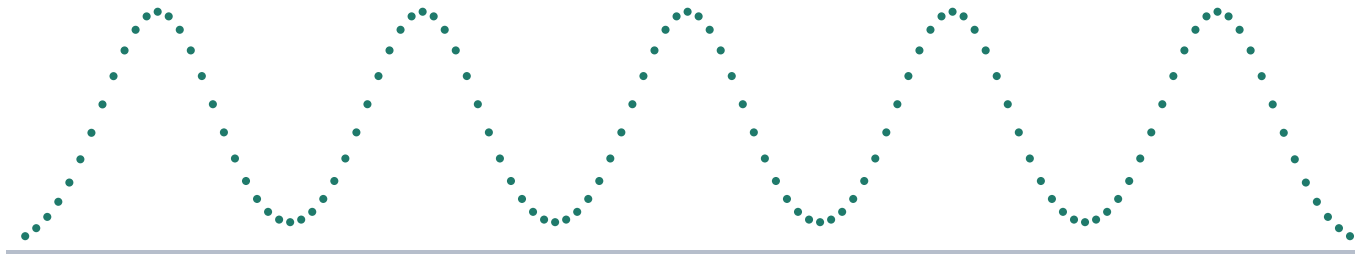
Mode collapse

The mechanism · 1:40–2:30

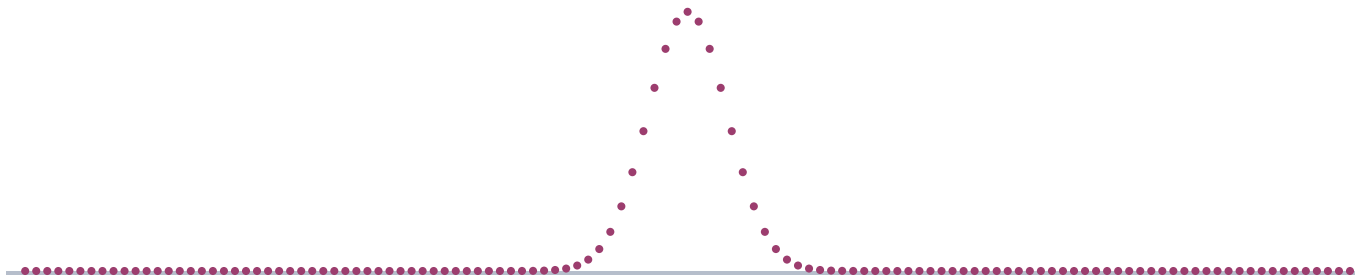
It has not broken. It is succeeding, narrowly and perfectly.



Five modes in the world, one in the model



p_{data} — the world, five modes



p_{g} — the model, one mode, perfectly convincing

A mode is a bump: a kind of thing the world regularly produces. Collapse is the loss of bumps, not the loss of quality.

Every sample it makes is good

The one thing it produces sits squarely on a real mode. Judged one output at a time, the model is flawless. There is no bad sample to point at.

Four fifths of the world is gone

Not badly rendered. Absent. The generator assigns essentially zero probability to regions the world visits constantly.

And no single output reveals it

Collapse is invisible at the level of the individual case and only appears in the distribution. You cannot detect it by looking harder at one example — you have to count.

Which is why you generated forty

One output tells you about quality. Forty tell you about coverage. The exercise was designed to make a distributional property visible to a person.

Why a working optimiser lands here

1

G finds one very convincing output

Under the non-saturating objective, G maximises $E[\log D(G(z))]$. If some output x^* currently fools D completely, mapping every noise vector z to x^* scores maximally. Variety is not rewarded, so variety is not produced.

2

D notices and starts rejecting x^*

D is still learning. Once all the fakes look like x^* , D learns that x^* is fake and $D(x^*)$ falls. G's reward collapses.

3

G moves to a different single output

It does not spread out. It relocates — abandoning x^* for some x^{**} that D has not yet caught. This is called mode hopping and it can continue indefinitely.

4

Nothing in the game punishes step three

A generator cycling between single outputs can keep V low forever without ever covering p_{data} . The equilibrium the theorem describes is never reached, and the objective cannot tell.

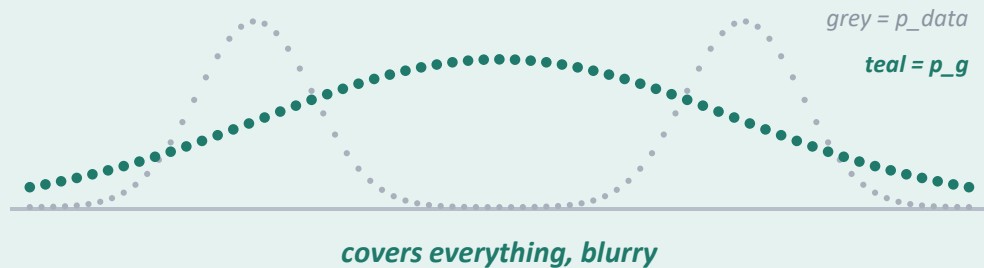
Every step is the optimiser doing exactly what last week's mathematics says it should. There is no malfunction anywhere in this list.

Two ways to be close, and only one of them keeps the modes

$$\text{KL}(p \parallel q) = \sum p(x) \log (p(x) / q(x))$$

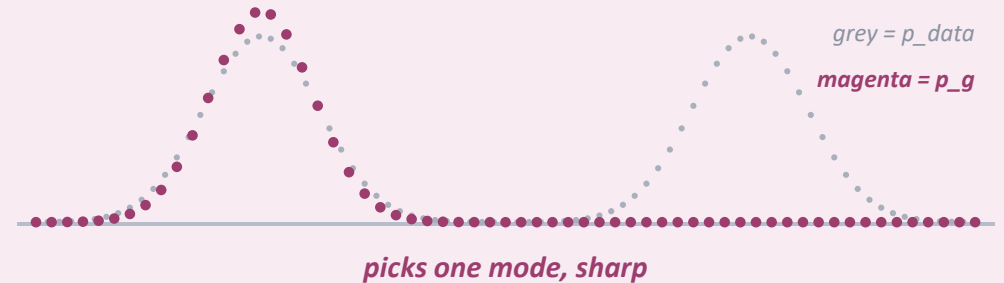
“how much you lose by using q when the truth is p ” — and it is not symmetric: $\text{KL}(p \parallel q) \neq \text{KL}(q \parallel p)$

$\text{KL}(p_{\text{data}} \parallel p_{\text{g}})$ — forward



The sum runs over the real world. Wherever p_{data} is large and p_{g} is near zero, the ratio explodes and the penalty is enormous. So the model is forced to put mass everywhere the world does. It covers all modes — and pays for it by smearing across the gaps.

$\text{KL}(p_{\text{g}} \parallel p_{\text{data}})$ — reverse



The sum runs over the model. Regions the model never visits cost nothing at all, because $p_{\text{g}} = 0$ there kills the term. The only penalty is for producing things the world does not. So: pick one real mode, sit on it, never stray.

Maximum-likelihood training minimises the forward direction, which is why those models blur. The GAN's practical objective behaves much more like the reverse direction — which is why these models drop modes and look sharp.

What the GAN objective is actually minimising

$$\text{JSD}(p \parallel q) = \frac{1}{2} \text{KL}(p \parallel m) + \frac{1}{2} \text{KL}(q \parallel m), \quad m = \frac{1}{2}(p + q)$$

“measure each distribution against their average, and split the difference”

Symmetric, and always finite

Unlike KL, $\text{JSD}(p \parallel q) = \text{JSD}(q \parallel p)$, and it never blows up to infinity even when one distribution assigns zero where the other does not. It is bounded above by $\log 2$.

It is what $C(G)$ reduces to

Substituting the optimal critic back into the game gives $C(G) = -\log 4 + 2 \cdot \text{JSD}(p_{\text{data}} \parallel p_g)$. Minimising the game is minimising a genuine distance between the world and the model.

So why does collapse happen anyway?

Because that identity assumes the optimal D . In practice D is a finite network, trained a few steps at a time, always slightly behind. G is optimising against an imperfect critic, and an imperfect critic can be satisfied narrowly.

The theory says the game measures a real distance. The practice says the game measures whatever this particular critic currently notices. Mode collapse is the difference between those two sentences, made visible.

Each one adds a term, and the term needs a weight

Minibatch discrimination

Let D look at a batch of generated samples together, rather than one at a time, so that “all of these are identical” becomes visible evidence of fakery. Diversity finally enters the objective — indirectly, through the critic.

Unrolled GANs

Let G optimise against where D is about to be, several steps ahead, rather than where D is now. This removes the incentive to chase a critic’s momentary blind spot.

Wasserstein GAN

Replace the JSD-flavoured objective with a distance that still gives useful gradients when the two distributions barely overlap, so G is not rewarded for hiding in a single place.

And the general shape

$L' = L + \lambda \cdot \text{diversity-ish-term}$.
Which is exactly last week’s impact regulariser: you must define the new term, choose λ , and accept whatever the new term leaves out.

The pattern, for the second week running

Nobody forgot about diversity. The repairs exist, they are ingenious, and several of them work well. What none of them does is make the problem go away, because each one requires somebody to write down what variety means, in a form a gradient can read, and weight it against quality. That specification problem is not solved by any of these — it is relocated.

PART FOUR

Measuring what died

Entropy, and effective modes · 2:40–3:10

A statistic you can compute by hand, on your own pages, in four minutes.



Entropy: how spread out a set of outputs is

$$H = - \sum p_i \log p_i$$

“for each kind of output, multiply how often it appeared by the log of how often it appeared, add them all up, flip the sign.”

p_i

the fraction of your forty outputs that fell into category i .

$\log$

natural log here, so H comes out in nats. Any base works; it just rescales.

the minus

logs of fractions are negative, so the minus makes H positive.

$H = 0$

every single output identical. Total collapse, measured.

H maximal

perfectly even spread across all the categories you found.

And a number you can interpret

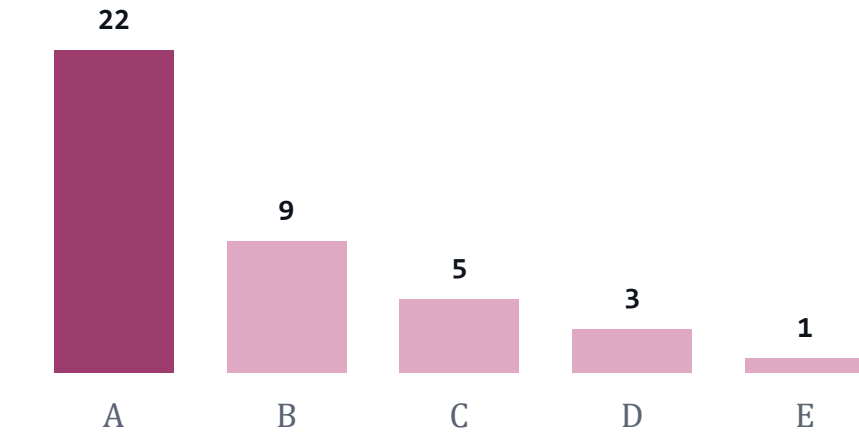
effective modes = $\exp(H)$

how many equally-likely kinds your set behaves like

Why $\exp(H)$ and not just H

H is in awkward units. Exponentiating turns it into a count: “this set of forty behaves like a fair choice among about three and a half things.” That is a sentence a person can argue with, which is what we want.

Forty outputs, five categories, one number



counts across 40 generated outputs

How to read it

Five distinct kinds appeared. But the distribution behaves like a fair choice among 3.36. If all five had been equally common, $\exp(H)$ would be exactly 5 — so roughly a third of the apparent variety is not really there.

category	count	p_i	$\log p_i$	$-p_i \log p_i$
A	22	0.550	-0.598	0.329
B	9	0.225	-1.492	0.336
C	5	0.125	-2.079	0.260
D	3	0.075	-2.590	0.194
E	1	0.025	-3.689	0.092

$$H = 1.211 \text{ nats} \quad \exp(H) = 3.36 \text{ effective modes}$$

Now do this with your own forty. Define your categories first, write the rule down, then count.

PART FIVE

Diagnose what died

Group work on your own outputs · 3:10–3:45

You now have the vocabulary you were denied at 0:20.



Diagnose what died

1 Pool and categorise

Put all your outputs together. Agree a category rule and write it down in one sentence. Then count, compute H , and compute $\exp(H)$ for the pooled set and for each individual's forty.

2 Find the silent absence

Name one thing that plainly belongs to the prompt's space and appears nowhere across all your outputs. Then say how you know it belongs — what makes it part of the space?

3 Locate the critic

What was playing the role of D here? Name it concretely. Training data, a preference model, a safety layer, your own prompt, your own sense of a good answer. There is more than one.

4 Test the mechanism

Does the §3 story actually fit what you see — a small set of outputs that reliably satisfy a critic? Or is something else going on? Argue for a verdict.

Report three minutes each: your category rule, your $\exp(H)$, and the one absence you are most confident about.

PART SIX

The same shape, larger

Bender et al. and Ruha Benjamin · 3:55–4:30

Two accounts of narrowing that do not need a generator to happen.



Large is not the same as diverse

1 Scale does not confer representativeness

A very large web corpus is not a sample of humanity. It overrepresents the already-connected, the already-published and the already-dominant. Size is being used as a proxy for coverage, and it is a bad one.

3 The snapshot is frozen

A corpus captures the world as it was during collection. Movements that change how people speak arrive after the freeze. The model keeps producing the earlier distribution — what the authors call value lock.

2 Filtering narrows it again

Cleaning pipelines built to remove harmful content also remove the language of the communities that reclaim slurs and discuss their own lives. The filter is a second critic, and it has preferences nobody wrote down.

4 And fluency reads as understanding

Output is coherent because we supply the coherence. Form is stitched together from observed patterns without reference to meaning, and a reader's interpretive generosity finishes the job.

What the paper is and is not arguing

It is an argument about data, documentation, cost and accountability, with concrete recommendations — datasheets, data statements, budgeting for curation. It is not a claim about an optimiser collapsing. Keep the two separate for the next forty minutes.

The New Jim Code, and the defence of the default

The central term

The New Jim Code: new technologies that reproduce existing inequities while being promoted and perceived as more objective or progressive than the discriminatory systems they replace. The claim to neutrality is part of the mechanism, not separate from it.

What her argument requires that ours does not

History. Institutions. Interests. Law. People who benefit and people who are harmed, named as such. A gradient has none of these, and a story about gradients cannot supply them. If you take one thing from this slide into the next block, take that.

Default discrimination

Systems reproduce inequity without anyone intending it, because they inherit a social terrain and treat its current arrangement as the baseline. The default is never neutral; it is a position that has stopped being visible as one.

Racism is not a glitch

Benjamin rejects the framing in which discriminatory output is a malfunction to be patched. A system producing unequal outcomes may be working precisely as built, on the materials it was given.

Which should sound familiar

It has not broken. It is succeeding. That sentence is the instructor's description of a collapsed generator and it is also, in a different register and with a different politics behind it, Benjamin's argument about technology.

And imagination is the intervention

Her abolitionist framing is not only diagnostic. What can be imagined constrains what gets built, so the contest over the imaginable is itself a site of action — which is a claim about narrowing that owes nothing to any optimiser.

PART SEVEN

Being suspicious of the metaphor

The instructor's real demand

A figure that fits too well is usually doing less work than it appears to.



Four things the metaphor carries, and four it does not

What transfers

- A narrow set of outputs that reliably satisfy a critic, produced to the exclusion of others.
- Excellence at the level of the individual case, coexisting with total failure at the level of coverage.
- Invisibility: no single output reveals it. You have to count.
- An objective with no term for variety, so variety is free to spend.

What does not

- A generator has one critic with a scalar output. A person has many, they conflict, and they can be argued with.
- p_{data} is well defined for a generator. “The diversity of the world” for a writer or a culture is not a distribution anyone can write down.
- Collapse is reversible by retraining. Institutional and cultural narrowing involves law, money, careers and harm, and does not reset.
- Nobody benefits from a collapsed GAN. Somebody usually benefits from a narrowed culture, and Benjamin insists we name them.

A metaphor you can audit like this is worth using. One you cannot is just a word borrowed for its authority.

What the metaphor can be used to do

It can make a political situation sound like a tuning problem

If a culture's narrowing is mode collapse, the implied remedy is a better objective and a larger λ . That framing has the enormous advantage of requiring nobody to give anything up, which is a reason to distrust it.

It can launder the critic

Saying a system "optimised for approval" makes approval sound like a natural force. Approval has a source. Someone is approving, for reasons, and benefits from what they approve of. The technical phrasing hides that grammar.

It can flatter the person using it

Reaching for a mechanism you understand to explain a situation you do not is a comfortable move and it produces a feeling of insight that is very hard to distinguish from the real thing.

But the alternative is not to refuse the analogy

The instructor's demand is harder than that: use the figure, and hold the mechanism in your hand while you use it, so that you can say at any moment which part of the claim is doing work and which part is decoration. That is the difference between a critical vocabulary and a fashionable one, and it is most of what this course is for.

What disappears when a system optimises for approval?

In the generator: everything the critic does not check

1

Not the things it does badly — the things it never attempts. The objective V contains no term for coverage, so coverage is available to be spent in exchange for more of what is measured. The model is not wrong. It is narrow, and being narrow costs it nothing.

In the corpus: everything the collection missed, then everything the filter cut

2

Bender's argument adds a stage the GAN story does not have. Before any optimisation, somebody decided what to gather and what to remove, and both decisions preferentially removed the already-marginal. The distribution was narrowed before training began.

In the world: whatever the default was already excluding

3

Benjamin's addition is that the baseline is not a neutral starting point but an inherited arrangement with beneficiaries. A system that reproduces it faithfully is not malfunctioning, and calling the result a collapse risks making it sound accidental when it is not.

Three different mechanisms. One shape. Saying which you mean, every time, is the skill.

Rewrite your warm-up

Return to the audience whose approval you learned, and what stopped coming out. Rewrite it using this session's vocabulary — name the critic, name the objective, name what was never in it.

Then do the harder half: state one respect in which mode collapse is the wrong word for what happened to you, and say what would be a better one.

A “Reasonable” Answer:

The second half. Anyone can apply a new term. The paragraph that identifies precisely where the term stops fitting, and proposes a replacement, is the one doing the work this unit was designed to teach.